Periodic Rotated Lattices in a Rectangular Box: Theory, Cell Construction and an Implementation Recipe

Square and triangular 2D lattices

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Abstract

We want atomic coordinates of a 2D crystal (square or triangular lattice) that is rotated by an angle θ with respect to a *rectangular* simulation box with periodic boundary conditions (PBC). This is only possible exactly for a discrete set of angles, labelled by a pair of coprime integers (m, n). This document (i) derives that condition, (ii) gives the box dimensions, the CSL index Σ and the number of atoms, (iii) gives an exact, integer-based rule to pick the atoms of one periodic cell without duplicates, and (iv) lists a step-by-step algorithm, edge cases and tests so that the construction can be coded directly.

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1 Overview and conventions

Goal. Produce N atom positions (x_k, y_k) inside the box $[0, L_x) \times [0, L_y)$ such that, when this box is repeated periodically in x and y, one obtains a perfect crystal (all nearest-neighbour distances equal to a, correct coordination number) whose lattice axes are rotated by θ relative to the box edges.

Key idea. Instead of rotating a lattice and hoping it fits, we do the reverse. Choose a lattice vector $\mathbf{t}$ of the *unrotated* crystal. Declare the x-axis of the box to be along $\mathbf{t}$. The box is then automatically commensurate with the crystal. The angle between the crystal axis $\mathbf{a}_1$ and $\mathbf{t}$ is the rotation angle θ, and the integer coordinates of $\mathbf{t}$ in the lattice basis are the pair (m, n).

Notation.

a	lattice constant (nearest-neighbour distance)
$\mathbf{a}_1, \mathbf{a}_2$	primitive vectors of the unrotated crystal
(m, n)	coprime integers, $m \geq n \geq 0$; $\mathbf{t} = m\mathbf{a}_1 + n\mathbf{a}_2$
θ	rotation angle of the crystal relative to the box
Σ	CSL index = area of the CSL cell / area of the primitive cell
L_x, L_y	box side lengths
N	number of atoms in the box

Frames and sign convention. We use two coordinate frames.

- **Crystal frame:** the lattice is unrotated, points are $i\mathbf{a}_1 + j\mathbf{a}_2$, $i, j \in \mathbb{Z}$.
- **Box frame:** the box edges are the x and y axes. A point is converted from the crystal frame to the box frame by a rotation by $-\theta$, where $\theta = \mathrm{atan2}(t_y, t_x)$ is the polar angle of $\mathbf{t}$.

So in the box frame the crystal appears rotated by $-\theta$. For $+\theta$ simply mirror $y \to -y$; both the square and the triangular lattice map onto themselves under this mirror (for the triangular lattice, $(a/2, -a\sqrt{3}/2) = \mathbf{a}_1 - \mathbf{a}_2$ is again a lattice vector), so the cell dimensions and atom counts are unchanged.

2 The problem: periodicity in a rectangular box

A rectangular PBC box is spanned by $\mathbf{L}_x = (L_x, 0)$ and $\mathbf{L}_y = (0, L_y)$. The system is a perfect crystal if and only if every atom at $\mathbf{P}$ has an identical atom at $\mathbf{P}+\mathbf{L}_x$ and at $\mathbf{P}+\mathbf{L}_y$. Equivalently:

> **Both $\mathbf{L}_x$ and $\mathbf{L}_y$ must be lattice vectors (translation vectors) of the rotated crystal.**

Because $\mathbf{L}_x \perp \mathbf{L}_y$, we need a pair of *perpendicular* lattice vectors. Such pairs exist only for special orientations. That is the whole problem.

2.1 Why only special angles work (direct derivation)

Take the square lattice rotated by θ, with basis $\mathbf{v}_1 = a(\cos\theta, \sin\theta)$ and $\mathbf{v}_2 = a(-\sin\theta, \cos\theta)$. We need integers i, j with

$$(L_x, 0) = i\,\mathbf{v}_1 + j\,\mathbf{v}_2.$$

The y-component gives $0 = a(i\sin\theta + j\cos\theta)$, hence $\tan\theta = -j/i$. A rational tangent is therefore necessary:

$$\tan\theta = \frac{n}{m} \qquad (m, n \in \mathbb{Z},\ \gcd(m, n) = 1). \tag{1}$$

Taking $i = m$, $j = -n$ gives the x-component $L_x = a(m\cos\theta + n\sin\theta) = a\sqrt{m^2+n^2}$, using $\cos\theta = m/\sqrt{m^2+n^2}$ and $\sin\theta = n/\sqrt{m^2+n^2}$. In the same way, $\mathbf{L}_y = n\mathbf{v}_1 + m\mathbf{v}_2 = (0,\, a\sqrt{m^2+n^2})$ is also a lattice vector, so no second condition arises for the square lattice: the perpendicular vector comes for free. This is the derivation that the box side $L = a\sqrt{\Sigma}$ (Eq. 5 below) relies on.

2.2 Which angles are therefore allowed?

- The allowed set is **dense but countable**: for any target angle there is an allowed angle arbitrarily close, but the price is a growing cell (large Σ).
- **Irrational** $\tan\theta$ (square) never works exactly, e.g. $30°$ for the square lattice, because $\tan 30° = 1/\sqrt{3}$.
- Angles outside the fundamental range follow by symmetry (Sec. 5).

3 Square lattice

3.1 Setup

Primitive vectors $\mathbf{a}_1 = (a, 0)$, $\mathbf{a}_2 = (0, a)$; one atom per primitive cell, area $A_0 = a^2$. For coprime (m, n) with $m \geq n \geq 0$ ($(1,0)$ is $\theta = 0$, $(1,1)$ is $45°$) define

$$\mathbf{t} = m\mathbf{a}_1 + n\mathbf{a}_2 = a(m,\, n), \qquad \mathbf{t}_\perp = -n\mathbf{a}_1 + m\mathbf{a}_2 = a(-n,\, m). \tag{2}$$

These two vectors are perpendicular, have equal length, and are both lattice vectors. They span the **CSL cell**, a square rotated by $\theta = \arctan(n/m)$ in the crystal frame.

3.2 Angle, index, box and atom count

$$\tan\theta = \frac{n}{m}, \tag{3}$$

$$\Sigma = m^2 + n^2, \tag{4}$$

$$L_x = L_y = a\sqrt{\Sigma}, \tag{5}$$

$$N = \Sigma. \tag{6}$$

Why Σ is the atom count. The area of the CSL cell is

$$A_{\mathrm{CSL}} = \left|\det \begin{pmatrix} ma & na \\ -na & ma \end{pmatrix}\right| = a^2(m^2+n^2),$$

and $\Sigma = A_{\mathrm{CSL}}/A_0 = m^2 + n^2$. A cell that is Σ times bigger than the 1-atom primitive cell contains Σ atoms. The box is the CSL cell itself, so $L_x \times L_y = a^2\Sigma$ is consistent with Eq. (5).

(m, n)	θ	$\Sigma = N$	L/a
$(1, 0)$	$0°$	1	1
$(1, 1)$	$45°$	2	$\sqrt{2} \approx 1.414$
$(2, 1)$	$26.565°$	5	$\sqrt{5} \approx 2.236$
$(3, 1)$	$18.435°$	10	3.162
$(3, 2)$	$33.690°$	13	3.606
$(4, 3)$	$36.870°$	25	5

4 Triangular lattice

4.1 Setup

Primitive vectors

$$\mathbf{a}_1 = (a, 0), \qquad \mathbf{a}_2 = \left(\tfrac{a}{2}, \tfrac{a\sqrt{3}}{2}\right),$$

primitive-cell area $A_0 = a^2\sqrt{3}/2$ (one atom). Let $\mathbf{t} = m\mathbf{a}_1 + n\mathbf{a}_2$, with (m, n) coprime and $m \geq n \geq 0$. In Cartesian components

$$\mathbf{t} = a\left(m + \tfrac{n}{2},\ \tfrac{\sqrt{3}}{2}n\right), \qquad |\mathbf{t}| = a\sqrt{m^2 + mn + n^2}.$$

4.2 Angle and CSL index

The polar angle of $\mathbf{t}$ gives the rotation angle:

$$\tan\theta = \frac{n\sqrt{3}}{2m + n}, \qquad \Sigma = m^2 + mn + n^2. \tag{7}$$

Derivation of Σ. The rhombic CSL cell is spanned by $\mathbf{t}$ and $\mathbf{t}$ rotated by 60°. A 60° rotation maps $\mathbf{a}_1 \to \mathbf{a}_2$ and $\mathbf{a}_2 \to \mathbf{a}_2 - \mathbf{a}_1$, so

$$\mathbf{L}_1 = m\mathbf{a}_1 + n\mathbf{a}_2, \qquad \mathbf{L}_2 = -n\mathbf{a}_1 + (m + n)\mathbf{a}_2.$$

Its area is $|\det \begin{pmatrix} m & n \\ -n & m+n \end{pmatrix}| A_0 = (m^2 + mn + n^2)A_0$, hence $\Sigma = m^2 + mn + n^2$ and this rhombic cell contains Σ atoms.

4.3 Rectangular box

The rhombic cell is useless for a rectangular PBC code, so we need a rectangle. Its x-side is $\mathbf{t}$:

$$L_x = |\mathbf{t}| = a\sqrt{\Sigma}.$$

The y-side must be a lattice vector perpendicular to $\mathbf{t}$. The vector $\mathbf{t}$ rotated by 90° is generally *not* a lattice vector, but $\sqrt{3}$ times it is. To see this, write lattice points as complex numbers $z = a(i + j\omega)$ with $\omega = e^{i\pi/3}$. These form a ring (the Eisenstein integers), and $i\sqrt{3} = 2\omega - 1$ belongs to it, so $i\sqrt{3}\,\mathbf{t}$ is again a lattice point. Therefore

$$L_x = a\sqrt{\Sigma}, \qquad L_y = a\sqrt{3\Sigma}, \qquad N = \frac{L_x L_y}{A_0} = 2\Sigma. \tag{8}$$

For $(m, n) = (2, 1)$: $\Sigma = 7$, $\theta = \arctan(\sqrt{3}/5) = 19.107°$, $N = 14$.

4.4 Refinement: a 3× smaller box when $3 \mid \Sigma$

If $m \equiv n \pmod 3$ then $\Sigma = m^2 + mn + n^2 \equiv (m - n)^2 \equiv 0 \pmod 3$. In that case $\mathbf{t}$ is divisible by $i\sqrt{3}$ inside the lattice, so a perpendicular lattice vector of length $|\mathbf{t}|/\sqrt{3}$ exists. The smallest rectangle is then

$$L_x = a\sqrt{\Sigma}, \qquad L_y = a\sqrt{\Sigma/3}, \qquad N = \frac{2\Sigma}{3} \qquad (\text{if } m \equiv n \bmod 3). \tag{9}$$

Example $(1, 1)$ ($\theta = 30°$): the box is $a\sqrt{3} \times a$ with 2 atoms, not $a\sqrt{3} \times 3a$ with 6 atoms. Both are valid periodic boxes; the formulas of Eq. (8) always work, Eq. (9) just gives the minimal one. It matters when the number of atoms must be kept small. Affected examples: $(1, 1)$, $(4, 1)$, $(5, 2)$, $(7, 1)$.

(m,n)	θ	Σ	L_x/a	L_y/a	N	note
$(1,0)$	$0°$	1	1	1.732	2	
$(1,1)$	$30°$	3	1.732	1	2	reduced
$(2,1)$	$19.107°$	7	2.646	4.583	14	
$(3,1)$	$13.898°$	13	3.606	6.245	26	
$(3,2)$	$23.413°$	19	4.359	7.550	38	
$(4,1)$	$10.893°$	21	4.583	2.646	14	reduced
$(5,2)$	$16.102°$	39	6.245	3.606	26	reduced

5 Choosing (m, n) and the fundamental angle range

Fundamental range. The square lattice has 4-fold symmetry and a mirror, so it is enough to treat $0 \le \theta \le 45°$ (i.e. $m \ge n \ge 0$). The triangular lattice has 6-fold symmetry and a mirror, so $0 \le \theta \le 30°$ is enough (again $m \ge n \ge 0$; $m = n$ gives $30°$). Any other angle is mapped to this range by $\theta \to \theta + k \cdot 90°$ (square) or $\theta + k \cdot 60°$ (triangular) and by $\theta \to -\theta$ and the mirrors about $45°$ resp. $30°$.

Restrict to coprime (m, n). Non-coprime pairs repeat an angle with a larger cell, e.g. $(2, 2)$ is $(1, 1)$ with Σ four times too large.

Given a target angle θ^*. Enumerate all coprime (m, n) with $\Sigma \le \Sigma_{\max}$, compute $\theta(m, n)$ from Eq. (3) or (7), and take the one minimizing $|\theta - \theta^*|$. State the achieved angle and the error to the user. Cell size grows roughly like $\Sigma \sim 1/\theta^2$ for small angles, so small angles need big cells.

When no small cell is close enough. Either accept a larger Σ, or slightly strain the crystal so that a small commensurate cell fits the target angle. If you strain, keep the strain well below what matters for the physics and report it.

6 Practical generation: the half-open cell

6.1 Why a half-open cell

Atoms sitting exactly on the left edge ($x = 0$) and on the right edge ($x = L_x$) are *periodic images of the same atom*. A cell that includes both edges contains duplicates; tiling it produces overlapping atoms (the "stacking" error). The fix is to keep atoms with

$$0 \le x < L_x, \qquad 0 \le y < L_y,$$

i.e. include left and bottom edges and exclude right and top edges. Then each atom of the infinite crystal is represented exactly once and $N = \Sigma$ (square) or 2Σ resp. $2\Sigma/3$ (triangular) as derived above. Figure 1 illustrates this for $\Sigma = 5$.

6.2 Exact integer selection rule (no floating-point tolerance needed)

Floating-point tests such as `fmod(x, Lx)` with atoms exactly on an edge are fragile: rounding can put an atom at $L_x - 10^{-16}$ instead of 0, giving duplicates or missing atoms. A robust alternative is to decide in the crystal frame with integers. A lattice point $p = i\mathbf{a}_1 + j\mathbf{a}_2$ has fractional box coordinates $u = (p \cdot \hat{\mathbf{t}})/L_x$ and $v = (p \cdot \hat{\mathbf{t}}_\perp)/L_y$ that are *rational with a known denominator* (Σ resp. 2Σ), so the half-open condition $0 \le u, v < 1$ becomes an integer inequality.

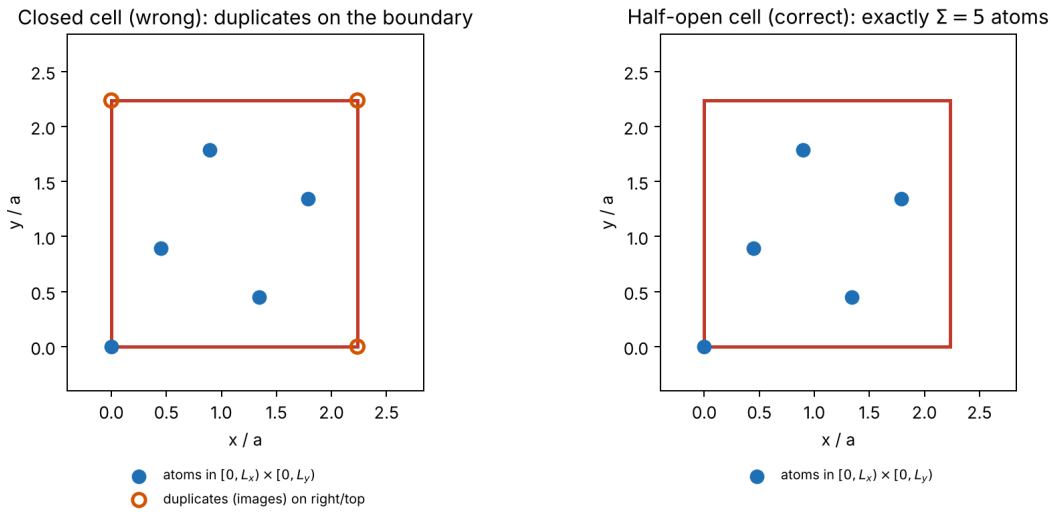


Figure 1: Square lattice, $\Sigma = 5$ ($\theta = 26.565°$). Left: a closed cell contains atoms on all four edges, with duplicates (orange circles). Right: the half-open cell contains exactly 5 atoms.

Square lattice.

$$U = im + jn, \qquad V = -in + jm, \qquad \text{keep } (i,j) \iff 0 \le U < \Sigma \text{ and } 0 \le V < \Sigma. \tag{10}$$

Here $u = U/\Sigma$ and $v = V/\Sigma$ are the fractional coordinates along L_x and L_y.

Triangular lattice (general box, Eq. (8)).

$$U = 2im+in+jm+2jn, \qquad V = jm-in, \qquad \text{keep } (i,j) \iff 0 \le U < 2\Sigma \text{ and } 0 \le V < 2\Sigma. \tag{11}$$

Triangular reduced box ($m \equiv n \bmod 3$, Eq. (9)): keep $0 \le U < 2\Sigma$ and $0 \le 3V < 2\Sigma$.

Where the formulas come from. With $\hat{\mathbf{t}} = \mathbf{t}/|\mathbf{t}|$ and $\hat{\mathbf{t}}_\perp$ the unit vector rotated by $+90°$, one has $u = (p \cdot \hat{\mathbf{t}})/L_x$ and $v = (p \cdot \hat{\mathbf{t}}_\perp)/L_y$. Inserting $p = i\mathbf{a}_1 + j\mathbf{a}_2$ and the box sides above gives exactly the integer numerators U, V and denominators Σ (square) resp. 2Σ (triangular).

Search range for (i,j). The corners of the cell in the crystal frame are 0, $\mathbf{t}$, $\mathbf{t}_\perp$ and $\mathbf{t} + \mathbf{t}_\perp$ (for the triangular box $\mathbf{t}_\perp$ has length $\sqrt{3}|\mathbf{t}|$). Looping over $|i|, |j| \le 3(m+n) + 3$ is safe and cheap; one may tighten it to a bounding box if needed.

6.3 Algorithm

```
INPUT : lattice in {square, triangular}, coprime integers m >= n >= 0, lattice
    constant a
OUTPUT: Lx, Ly, array of N atom positions (x_k, y_k) in [0,Lx) x [0,Ly)

1. Sigma = m^2+n^2                          (square)
           m^2+m*n+n^2                      (triangular)
2. theta = atan2(n, m)                      (square)
           atan2(n*sqrt(3), 2*m+n)          (triangular)
3. Box:
   square      : Lx = Ly = a*sqrt(Sigma),                N = Sigma
   triangular  : Lx = a*sqrt(Sigma)
                 if (m-n) % 3 == 0: Ly = a*sqrt(Sigma/3), N = 2*Sigma/3   (reduced)
```

```
                else             : Ly = a*sqrt(3*Sigma), N = 2*Sigma
4. For all integer (i,j) with |i|,|j| <= 3*(m+n)+3:
     compute (U,V) and test the integer inequalities of the previous section
     if kept:
        crystal-frame position  p = i*a1 + j*a2
           square:     p = a*(i, j)
           triangular: p = a*(i + j/2, j*sqrt(3)/2)
        box-frame position (rotate by -theta):
           x =  cos(theta)*p.x + sin(theta)*p.y
           y = -sin(theta)*p.x + cos(theta)*p.y
        append (x,y)
5. Assert len(atoms) == N.
6. Replicate by integer shifts (k*Lx, l*Ly) for a larger sample, if wanted.
```

Atoms that sit at $x \approx 0$ or $y \approx 0$ after rotation may come out as $\pm 10^{-16}$. Clamp tiny negative values to 0 after the selection (the selection itself is exact, so this is cosmetic). For a periodic output, wrap with $x \leftarrow x - L_x \lfloor x/L_x + \varepsilon \rfloor$ with $\varepsilon \approx 10^{-9}$ rather than a bare `fmod`.

6.4 Tests the code must pass

1. **Atom count:** $N = \Sigma$ (square), 2Σ or $2\Sigma/3$ (triangular).
2. **No duplicates:** all pairwise minimum-image distances > 0.
3. **Nearest-neighbour distance** (minimum image, using L_x, L_y) equals a for every atom.
4. **Coordination:** every atom has exactly 4 (square) or 6 (triangular) neighbours at distance a under PBC. For very small cells, count neighbours including periodic images of the same atom, otherwise $\Sigma = 1, 2$ look wrong.
5. **Translation invariance:** the point set shifted by $(L_x, 0)$ and $(0, L_y)$ and wrapped into the box reproduces the same set.
6. **Angle:** the bond direction in the box frame makes the angle $\pm\theta$ (mod the symmetry of the lattice) with the x-axis.

Tests 1, 3 and 4 were run with a reference implementation on every coprime pair with $m \leq 7$ and $\Sigma \leq 80$ for both lattices, and all passed.

7 Worked examples and figures

7.1 Square lattice, $\Sigma = 5$ ($\theta = 26.565°$)

$(m, n) = (2, 1)$, $\mathbf{t} = a(2, 1)$, $\mathbf{t}_\perp = a(-1, 2)$. The box is $a\sqrt{5} \times a\sqrt{5} \approx 2.236a$ and holds 5 atoms. Selection rule (10): $U = 2i+j$, $V = -i+2j$; the five admitted pairs are $(0, 0), (0, 1), (1, 1), (0, 2), (1, 2)$, shown in blue in Fig. 2.

7.2 Triangular lattice, $\Sigma = 7$ ($\theta = 19.107°$)

$(m, n) = (2, 1)$. The box is $a\sqrt{7} \times a\sqrt{21} \approx 2.646a \times 4.583a$ with 14 atoms (Fig. 3).

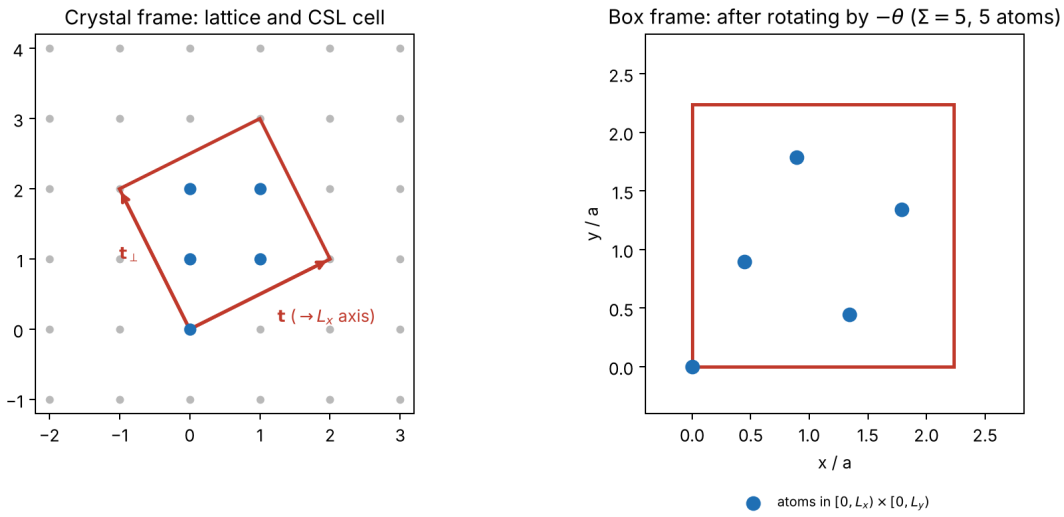


Figure 2: Square lattice, $\Sigma = 5$. Left: in the crystal frame the CSL cell (red) is a tilted square spanned by $\mathbf{t}$ and $\mathbf{t}_\perp$; blue dots are the admitted atoms. Right: the same atoms in the box frame after rotation by $-\theta$.

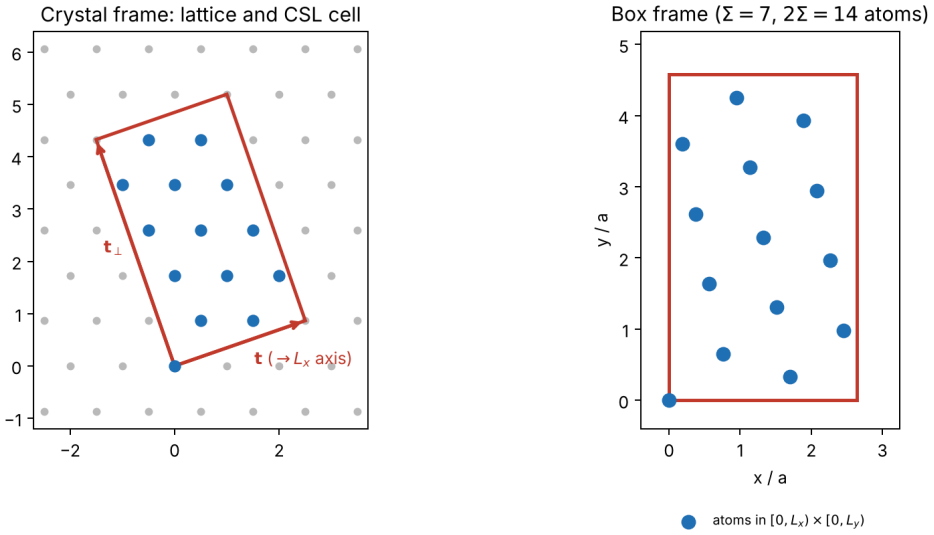


Figure 3: Triangular lattice, $\Sigma = 7$. The CSL rectangle has aspect ratio $\sqrt{3}$ and contains $2\Sigma = 14$ atoms.

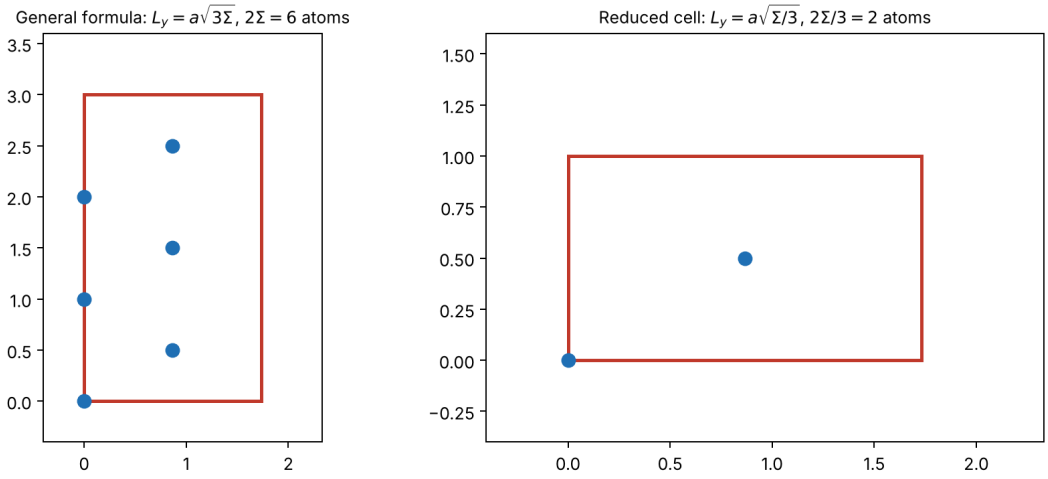


Figure 4: Both boxes are periodic; the reduced one is three times smaller (Sec. 4.4).

7.3 Triangular lattice, $\Sigma = 3$ (30°): general vs. reduced box

8 Summary and pitfalls

	Square	Triangular
Angle	$\tan\theta = n/m$	$\tan\theta = \dfrac{n\sqrt{3}}{2m+n}$
Σ	$m^2 + n^2$	$m^2 + mn + n^2$
L_x	$a\sqrt{\Sigma}$	$a\sqrt{\Sigma}$
L_y	$a\sqrt{\Sigma}$	$a\sqrt{3\Sigma}$ ($a\sqrt{\Sigma/3}$ if $m \equiv n \bmod 3$)
N	Σ	2Σ ($2\Sigma/3$ if $m \equiv n \bmod 3$)
Fundamental range	0–45°	0–30°

- Not every angle is possible: $\tan\theta$ must be rational (square) or of the form $n\sqrt{3}/(2m+n)$ (triangular). E.g. 30° is impossible for the square lattice; 15° and 45° are impossible for the triangular lattice.
- Always use coprime (m, n).
- Always use the half-open cell $[0, L_x) \times [0, L_y)$.
- The rotation direction is a convention; check which sign the downstream code expects.
- A target angle that is not allowed requires either a larger Σ (closer angle) or a small, documented strain.